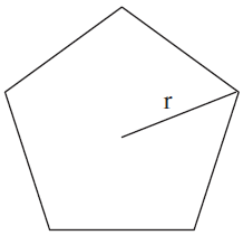
Sheet (2)

(1) The side length of a regular pentagon is expressed as $A = \frac{5}{2} r^2 . \sin\left(\frac{2\pi}{5}\right)$ where r is the length from the center of a pentagon to a vertex



Write a Python program that prompts the user to enter the length from the center of a pentagon to a vertex and computes the area of the pentagon, as shown in the given figure.

(2) The area of a pentagon can be computed using the following formula $A = \frac{5 \times s^2}{4 \times \tan(\pi/5)}$ where s is the length of a side. Write a program that prompts the user to enter the side of a pentagon and displays the area.

(3) A regular polygon is an n-sided polygon in which all sides are of the same length and all angles have the same degree (i.e., the polygon is both equilateral and equiangular). The formula for computing the area of a regular polygon is expressed as $A = \frac{n \times s^2}{4 \times \tan(\pi/n)}$ where s is the length of a side.

Write a program that prompts the user to enter the number of sides and their length of a regular polygon and displays its area.

(4) Write a Python program that prompts the user to enter the coordinates of the three vertices, (x_1 , y_1) , (x_2 , y_2) , and (x_3 , y_3) of a triangle. The program should then calculate and display the length of each side and the inner angles of the tringle.
Not that

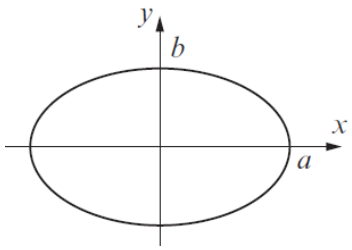
- The distance between any two points (x_n , y_n) , and (x_{n+1} , y_{n+1}) is calculated as $\sqrt{(x_{n+1} - x_n)^2 + (y_{n+1} - y_n)^2}$
- If the sides of a triangle are a, b and c then its corresponding inner angles A , B and C can be expressed as;

$$A = \cos^{-1}\left(\frac{a^2 - b^2 - c^2}{-2bc}\right),\ B = \cos^{-1}\left(\frac{b^2 - a^2 - c^2}{-2ac}\right),\ C = \cos^{-1}\left(\frac{c^2 - b^2 - a^2}{-2ab}\right)$$

(5) The perimeter of an ellipse can be approximated by

$$p = \pi\left[3(a+b) - \sqrt{(3a+b)(a+3b)}\right]$$

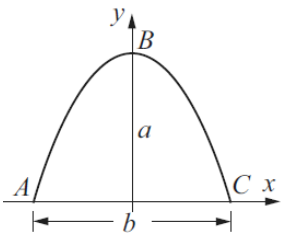
Calculate the perimeter of an ellipse with $a = 18\ in.$ and $b = 7\ in.$



(6) The arc length of a segment of a parabola *ABC* of an ellipse with semi-minor axes a and b is given approximately by:

$$L_{ABC} = \frac{1}{2}\sqrt{b^2+16a^2} + \frac{b^2}{8a}\ln\left(\frac{4a+\sqrt{b^2+16a^2}}{b}\right)$$

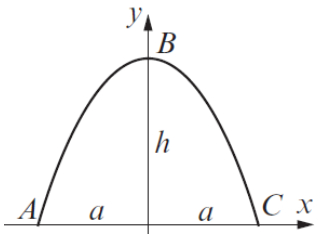
Determine L_{ABC} if a=11 in. and b=9 in.



(7) The arc length of a segment of a parabola *ABC* is given by:

$$L_{ABC} = \sqrt{a^2+4h^2} + \frac{a^2}{2h}\ln\left(\frac{2h}{a} + \sqrt{\left(\frac{2h}{a}\right)^2+1}\right)$$

Determine L_{ABC} if $a = 8$ in. and $h = 13$ in.



(8) The gravitational force between two bodies is expressed as

$$F = G \times \frac{m_1 \times m_2}{d} \quad \text{Newtons}$$

where $G = 6.6743 \times 10^{-11}\ N.m^2/kg^2$ is the gravitational constant.
m_1 and m_2 are the masses of two bodies in kgs and
d is the distance between them in m.

Write a Python program, which defines gravitational constant and prompts the user to enter the masses of the two bodies and the distance between them, then calculates and prints the gravitational force.

Typical data: Mass of the earth $= 5.97 \times 10^{24}\ kg$ - Mass of the moon $= 7.3 \times 10^{22}\ kg$ – Distance between Earth and Moon $3.84 \times 10^{8}\ m$ – Average gravitational force between the Moon and Earth $1.973 \times 10^{20}\ N$

(9) Let (x_1, y_1) and (x_2, y_2) be the geographical latitude and longitude of two points. The great circle distance between the two points can be computed using the following formula:

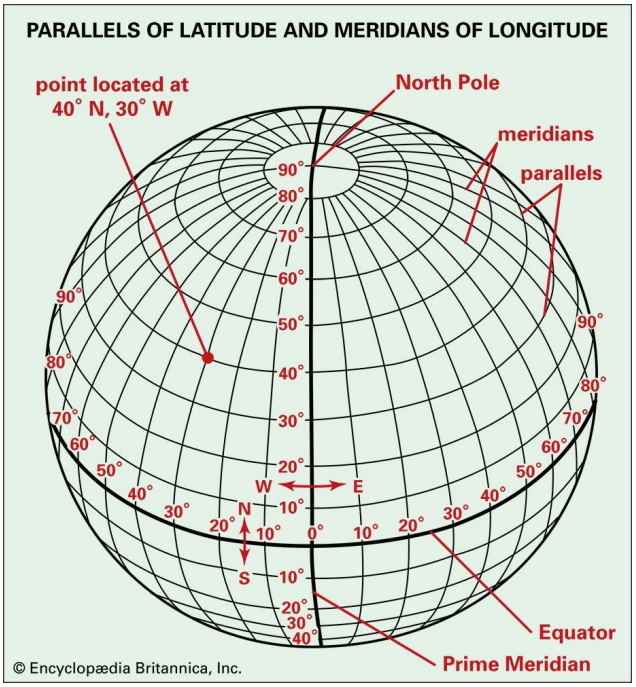
$$d = R \times cos^{-1}[sin(x_1) \times sin(x_2) + cos(x_1) \times cos(x_2) \times cos(y_1 - y_2)]$$

Write a Python program that prompts the user to enter the latitude and longitude of two points on the earth in degrees and displays its great circle distance. The average earth radius (R) is 6,371.01 km.

Run your program to calculate the distance between Cairo, Egypt and London, United Kingdom

Cairo Airport: latitude $30.116° \; North$ and Longitude $31.417° \; East$

London Heathrow Airport $51.468° \; North$ and Longitude $0.459° \; West$



(10) Write a program that receives an ASCII code (an integer between 32 and 126) and displays its character. For example, if the user enters 97, the program displays the character a.

(11) Write a program that receives a printable character and displays its corresponding decimal ASCII code character. For example, if the user enters character @, the program displays the decimal code 64.